Josh Ryther

Full-Stack Software Engineer · AI-Forward Development

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SUMMARY

Full-stack engineer with 4 years building secure software that delivers **millions of high-stakes tests a year**, from accessible TypeScript, Vue, and React interfaces to the services and REST APIs behind them. AI-forward: I pair coding agents with custom skills, automated tests, and careful review to ship faster with fewer bugs.

~50% faster development · ~75% fewer bugs · Millions of exams a year

SKILLS

Languages: JavaScript, TypeScript, HTML, CSS, Java, SQL
Frontend: Vue 3, React (Redux), Tailwind CSS, WCAG 2.1/2.2 AA, jQuery
Backend: Node.js, AWS Lambda, REST APIs, Spring, DynamoDB, SQL, MongoDB
AI & Testing: AI coding agents, custom agent skills, MCP, Playwright, Jest, Vitest, ESLint, Jenkins CI, Git

EXPERIENCE

Software Engineer — Pearson Education

Remote · Sept 2022 – Present
TestNav, the student testing app behind millions of secure exams a year in Chrome, Firefox, and Safari on desktop, iOS, and ChromeOS

- **~50% faster development** with an AI-forward workflow: custom agent skills, a Playwright MCP skill agents use to verify their own changes, and a tool that runs multiple coding agents on one project at once.
- **~75% fewer bugs** in new features after I introduced Playwright end-to-end testing to a codebase that had none.
- **Modernized the project's views** from legacy JavaScript as reusable TypeScript components in React (Redux) and Vue.
- **WCAG 2.1 AA compliance** reached with accessibility consultants; my reusable screen reader component cut accessibility work **from 3+ days to 1**.
- **~200 ms cached responses** from an AI-powered backend feature I built end to end in TypeScript on AWS Lambda, with ESLint and unit tests.
- **50–75% faster iOS development** after I traced and fixed the team's broken local debugging to a new OS-level requirement.
- **Deprecated core dependency removed** from a complex legacy feature, adding features with full backward compatibility.
- **50–75% less development time** per test item with reusable interactive content templates I designed and QA'ed.

Certified Pharmacy Technician — Ro (startup), Providence Health Plan, Walgreens

Beaverton, OR · 2013 – 2022

- Accuracy-critical prescription and prior-authorization work; mentored new hires while completing a CS degree.

PROJECT

AI-Powered Portfolio — joshryther.com

Vue 3, TypeScript, WebAssembly (llama.cpp), Vitest

- **In-browser AI chat** with on-device semantic search (Google EmbeddingGemma on the visitor's CPU), so no question leaves their device.
- **Built to production standards:** WCAG 2.2 AA (0 axe-core violations), Playwright end-to-end tests on desktop and mobile in CI with Dependabot auto-merge, English and Spanish, light and dark themes, and a frame-rate-independent canvas mini-game.

EDUCATION

B.S. Computer Science, Southern New Hampshire University, 2022 · **B.S. Economics**, University of Oregon, 2014